

Model Risk & Validation Report

Benchmark, mock and outcome testing of the trade-idea models, structured after the Federal Reserve's SR 11-7 guidance on model risk management. Test run: September 24, 2026. Data window: 2026-08-24 to 2026-09-23.

1. Executive summary

We tested 16 model behaviours with synthetic inputs of known answer (15 passed), analysed **756 resolved trade ideas** (274 reached target or stop), benchmarked every idea against its own random-walk break-even rate, ran a walk-forward benchmark of the bottom-reversal detector over **37,611 stock-sessions**, and reviewed the live paper book (27 closed trades).

Finding	Result	Severity
Bottom-reversal detector beats the market	+0.68% over 10 sessions vs universe ($t = 3.3$)	Validated
Legacy SPX-session producer loses money	12.1% hit vs 40.2% break-even; -0.51R per idea ($t = -6.7$)	Critical
Conviction score does not rank outcomes	AUC 0.46, Spearman rho 0.03 ($n = 337$)	Critical
Stated confidence is miscalibrated	Ideas stated ~89% confident hit ~32%	Critical
Legacy book overall below random walk	22.3% hit vs 33.0% break-even ($z = -3.9$); -0.24R	High
Shorts weaker than longs historically	-0.40R vs -0.20R	Monitor
Fast V-recoveries never qualify	Stop anchors on the capitulation bar; risk exceeds the 10% cap	Medium
Live paper book	\$429 net, profit factor 1.04, max drawdown -\$6,298	Monitor

Scope caveat. The resolved history is dominated by the producers that ran before this week's rebuild (quant, SPX-session, GEX, early market scanner). The rebuilt producers — aggressor-tape, leader-swing, index-swing, premium-discount and crypto-transmission — have no resolved outcomes yet; only the bottom-reversal detector is validated here, by walk-forward replay. The window is one month of one market regime: every figure is evidence, not proof.

2. Framework and method

Model inventory. Ten production models were in scope: the conviction engine (layered scoring), aggressor-tape scanner (long and short), bottom-reversal detector, leader-swing and index-swing detectors, premium-discount scan, crypto-transmission promoter, context target ladder, contract picker / live repricer, and the discipline gates (short policy, leveraged-ETF block, tape contradiction).

Conceptual soundness. The production functions were called directly with synthetic price series whose correct answer is known in advance, including negative controls that must not fire. Probability math was checked against published normal-distribution table values.

Outcomes analysis. For every resolved idea, the realized move was expressed in R (the distance to the stop). Ideas that expired without an exit price carry no R; win rates use only ideas that reached target or stop.

Benchmarking. Each idea is compared with the probability a driftless random walk reaches its target before its stop: $\text{risk} \div (\text{risk} + \text{reward})$. Summing these gives the number of wins chance alone would produce; the z-score measures how far the realized count sits from it. The detector is benchmarked walk-forward: at each historical session, it sees only bars up to that day, and its fires' forward returns are compared with every stock's forward return over the same sessions.

Discrimination and calibration. Whether a higher conviction score predicts a better outcome is measured with the AUC (probability a winner scored higher than a loser; 0.5 = no skill) and Spearman rank correlation. Calibration compares each idea's stated confidence with its realized hit rate.

Stability. The resolved set is split chronologically in half; a real edge should appear in both halves.

3. Conceptual soundness — mock tests

ID	Test	Expected	Actual	
A1	Normal CDF vs published table values (5 points)	max error < 1e-4	max error 4.48e-5	PASS
A2	CDF symmetry $\Phi(x)+\Phi(-x)=1$	< 1e-7	0.00e+0	PASS
A3	Expected move 1σ (spot 100, IV 40%, 9 DTE)	\$6.28	\$6.28	PASS
A4	Driftless touch odds at 1σ (reflection principle)	$\approx 31.7\%$	31.7%	PASS
A5	Planted V-recovery (–20% then 55% retrace in 4 sessions)	fires as v_recovery	no fire	FAIL
A5b	Planted V-recovery, moderate pace (–14% then 54% retrace over 5 sessions)	fires as v_recovery	v_recovery	PASS
A6	Planted higher-lows base (–16% then ascending floors)	fires as higher_lows_base	higher_lows_base	PASS
A7	NEGATIVE CONTROL: steady uptrend, no bottom	no fire	no fire	PASS
A8	NEGATIVE CONTROL: ongoing crash, no recovery	no fire	no fire	PASS
A11	Planted 2% pullback inside a rising trend	fires	fires (2.5% pullback)	PASS
A12	NEGATIVE CONTROL: downtrend (no counter-trend longs)	no fire	no fire	PASS
A13	NEGATIVE CONTROL: at the high, no discount	no fire	no fire	PASS
A14	Short with measured evidence, no news event (policy 2026-09-24)	allowed	allowed	PASS
A15	BTC-proxy short while bitcoin is rising	blocked	blocked	PASS
A16	BTC-proxy short while bitcoin is falling	allowed	allowed	PASS
A17	Leveraged/inverse wrappers blocked, sector ETFs allowed	TQQQ/SOXL/UCO/SCO blocked; SPY/SMH/QQQ allowed	as expected	PASS

15 of 16 passed. The failure (A5) is a genuine design finding, not a test defect: on a fast V-shaped recovery the stop anchors on the capitulation bar, so risk exceeds the 10% cap and the detector rejects the setup. The sharpest recoveries — often the strongest — can therefore never qualify. A moderate-pace V (A5b) fires correctly. Negative controls (uptrend, crash, downtrend, no-discount) all correctly stayed silent.

4. Outcomes analysis and benchmarking

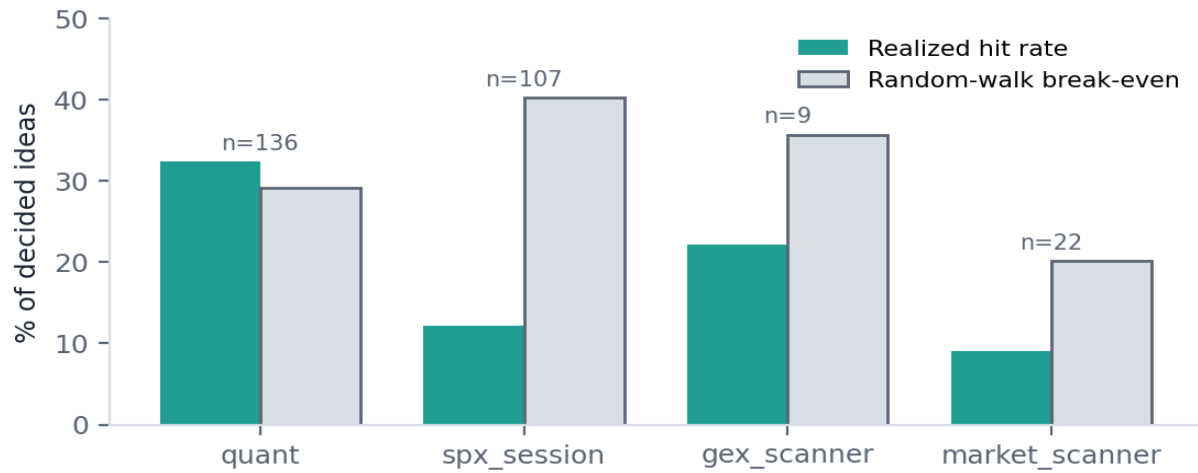


Figure 1. Realized hit rate against each idea's random-walk break-even rate, by producer. Bars below grey lose to chance.

Producer	Ideas	Decided	Hit rate	Break-even z	E[R]	t	PF	
quant	393	136	32.4%	29.2%	0.85	-0.165R	-1.81	0.72
spx_session	190	107	12.1%	40.2%	-5.96	-0.511R	-6.67	0.28
gex_scanner	28	9	22.2%	35.7%	-0.87	-0.191R	-0.90	0.52
market_scanner	143	22	9.1%	20.1%	-1.34	-0.027R	-0.24	0.94
All	756	274	22.3%	33.0%	-3.91	-0.245R	-4.66	0.58

The SPX-session producer is the clearest failure: it reached target on 12.1% of decided ideas when chance alone predicts 40.2% ($z = -6.0$), losing 0.51R per idea. The quant producer beats its break-even on hits ($z = 0.85$) but expired trades pull its expectancy below zero — its exits, not its entries, are the weakness.

Long versus short

Side	Ideas	Hit rate	Break-even	Expectancy	t-stat
Long	606	24.6%	30.2%	-0.199R	-3.31
Short	150	16.9%	39.4%	-0.397R	-3.69

Historical shorts (under the earlier event-required policy) underperformed longs. The new policy lets shorts in on measured evidence alone; their results should be reported separately for the first 60 trading days.

5. Discrimination and calibration

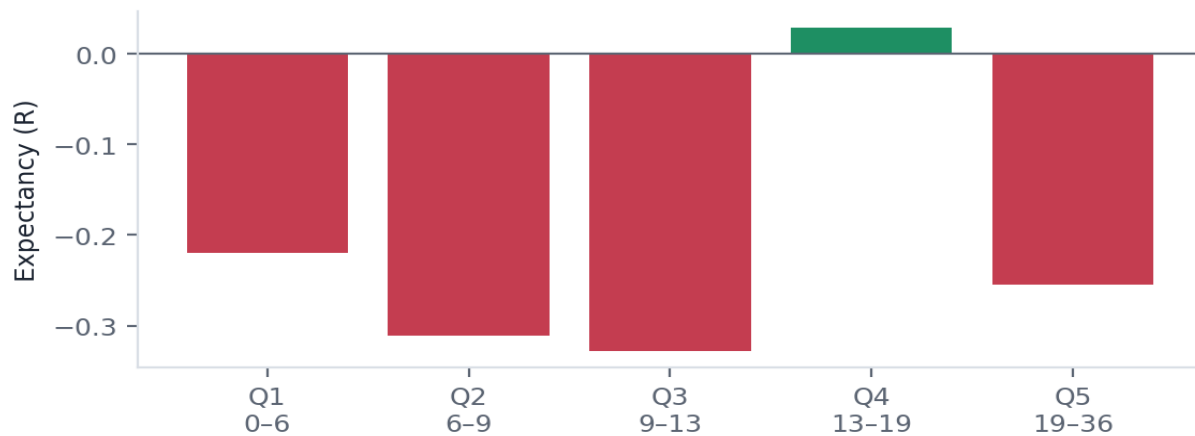


Figure 2. Expectancy by conviction-score quintile (n = 337). A working score rises left to right; this one does not. AUC 0.46 and Spearman rho 0.03 both indicate no ranking skill.

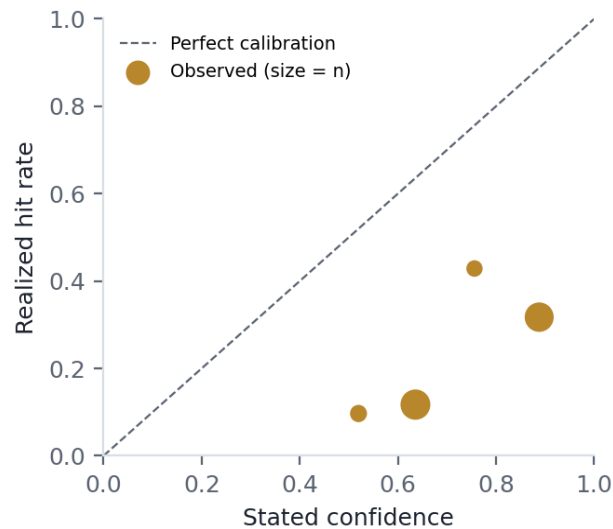


Figure 3. Stated confidence against realized hit rate. Points far below the dashed line are over-confident: ideas stated ~89% confident reached target ~32% of the time.

Confidence band	Ideas	Mean stated	Realized hit rate
0–60	31	52.0%	9.7%
60–70	111	63.6%	11.7%
70–80	28	75.6%	42.9%
80–100	104	88.8%	31.7%

6. Detector benchmark (walk-forward)

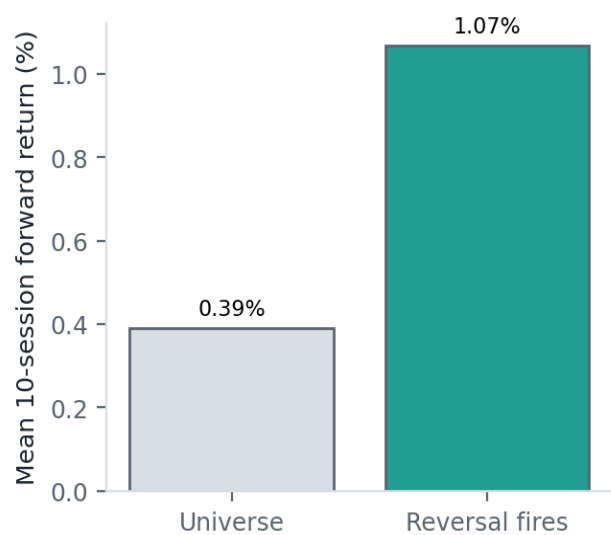


Figure 4. Bottom-reversal fires (score 70+, n = 2,021) against every liquid stock over the same sessions (n = 37,611).

Metric	Reversal fires	Universe
Mean 10-session return	1.07%	0.39%
Share positive	53.7%	47.3%
Median return	0.52%	—
Excess (t-stat)	+0.68% (t = 3.32)	

The detector sees only bars up to each test date (no look-ahead). Its excess return is statistically significant at the 1% level. Overlapping 10-session windows inflate the effective sample; the t-stat should be read as indicative, and the edge confirmed on live outcomes.

7. Stability and the live paper book

Half	Hit rate	Break-even	Expectancy	t-stat
First half	27.3%	35.0%	-0.242R	-3.28
Second half	10.0%	28.1%	-0.247R	-3.30

Split at 2026-08-26. The legacy book's negative expectancy is stable across both halves — a structural problem, not a bad week.



Figure 5. Live paper book, 27 closed trades: net \$429, win rate 37.0%, average win \$1,036 vs average loss \$584, profit factor 1.04, maximum drawdown -\$6,298.

8. Work completed during this review

- Replaced carried-forward contract values with live repricing: delta, IV, expected move, target odds and premium drift recomputed every two minutes.
- Fixed seven stale-value hazards found by code audit, including position sizing from publish-time premium and an options-chain cache that served stale data indefinitely.
- Added coherence gates: gamma-pin geometry corrected, opposing aggressor flow now subtracts from conviction, tape flips retire the opposite-side card.
- Replaced formula targets with a context target ladder (structure, dealer walls, prior swings) and a path-to-structure scoring layer.
- Brought shorts to parity with longs under a measured-evidence standard; blocked leveraged/inverse wrappers at the shared gate.
- Corrected the sector-rotation baseline (week-old reference was being shown as today's move) and the missing latest candle on every chart.
- Imported the Supabase backup history and moved the platform to an owned database and server.

9. Recommendations

#	Action	Why	Priority
1	Suspend the SPX-session producer pending redesign	Loses 0.51R per idea; far below chance, in both halves	Immediate
2	Stop presenting 'confidence' as a probability	89% stated vs 32% realized; recalibrate on outcomes or relabel as a rank	Immediate
3	Re-fit conviction layer weights on outcome data	Current score has no ranking skill (AUC 0.46)	High
4	Report shorts separately for 60 trading days	Historical shorts underperformed; new policy needs its own record	High
5	Widen the V-recovery stop logic	Fast V-recoveries are structurally excluded (test A5)	Medium
6	Re-run this harness monthly on new outcomes	Only the reversal detector has out-of-sample evidence so far	Ongoing

Reproduce: `npx tsx research/model-validation.ts` then `python3 research/build-validation-report.py`. Raw results: `research/validation-results.json`. QuantEdge is an educational research platform; nothing here is investment advice.